

Bluetooth Operated LED Desk Lamp

T.K. Hareendran

This LED desk lamp can be switched on or off from a distance using Bluetooth. You can even regulate the lamp's brightness through a slider on the screen of your Android smartphone.

The key hardware required for this little project is an Arduino Uno micro-controller board that enables you to control the lamp. Wiring diagram for the same is shown in Fig. 1.

Working

Regulated 5V DC from the Arduino Uno's onboard linear voltage regulator powers the HC-05 compatible Bluetooth module. The module com-

municates with the Arduino through its Tx and Rx lines that are connected to pins 2 and 3 of the Arduino board, respectively.

Pin 9 of the Arduino drives the white LED light source LED1 through the S8050 high-current transistor T1. The 1-kilo-ohm resistor R1 is the base resistor of driver transistor T1.

When the Arduino board is powered through its DC input jack (see Fig. 2), the DC voltage appears at its Vin pin through M7 (1N4007) silicon diode that is connected in series with the positive rail. This voltage is used to power the white LED light source. There is an inherent voltage drop of around 700mV, though.

Likewise, the 700mA (I_C max) small-signal transistor S8050 has a $V_{CE(SAT)}$ close to 500mV ($I_C = 500mA$, $I_B = 50mA$). Therefore, the white LED light source receives

Hardware

- Arduino Uno (Rev3)
- HC-05 compatible Bluetooth module
- S8050 NPN transistor (T1)
- White LED light (LED1)
- 1-kilo-ohm resistor (R1)
- 12V DC (<500mA) white LED module
- 230V AC to 12V DC (>750mA) SMPS power supply
- Connector for Bluetooth module (CON1)

a slightly lower operating voltage than the input 12V DC. However, this is not a big issue because almost all 12V DC white LED modules work well with 9V DC onwards.

Software

The Arduino sketch (code) given below is based on RemoteXY Library. Its latest version can be downloaded from <https://remotexy.com/en/library> and installed in the Arduino IDE. The steps for installation can be found in the link <https://remotexy.com/en/help/how-install-library/>

```
/* Code
* Project: Bluetooth LED Desk Lamp v1
* Hardware: Arduino Uno & HC-05 Compatible Bluetooth Module
* Android App: RemoteXY (http://remotexy.com/en/download/)
* Arduino Library: RemoteXY 2.4.6
```

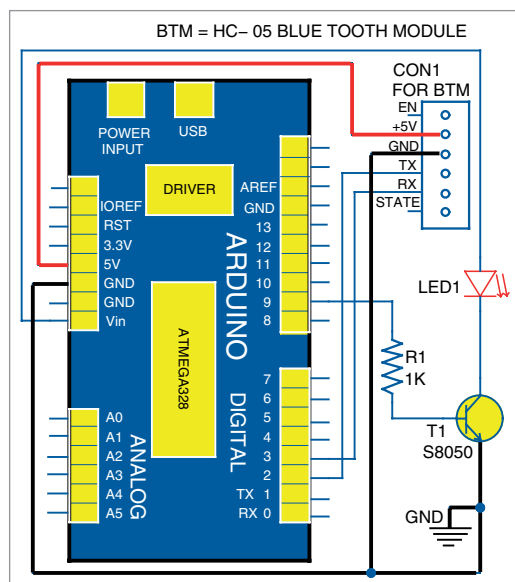


Fig. 1: Wiring diagram for the LED desk lamp

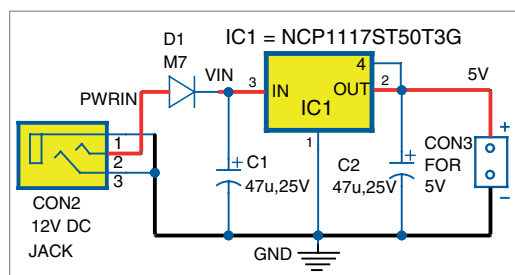


Fig. 2: Internal power supply circuit of Arduino

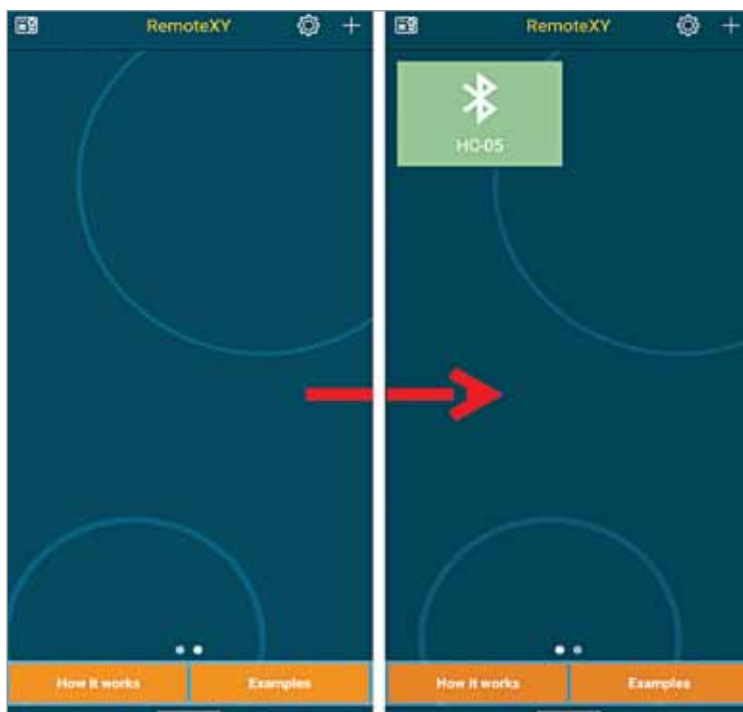


Fig. 3: RemoteXY Android app